Track 2 — Drug Repurposing Report

Rare Disease, Real Kid: MVA Hackathon 2026 · Proband `PROBAND01`

> **Read this first.** This is a research hypothesis prepared for a hackathon. It is **not medical advice and not a treatment recommendation.** Every agent discussed would be off-label for this child, and any decision belongs to the child's clinical team and family. Nothing here uses information beyond what the challenge provided.

1. Summary — the proposal

`PROBAND01` has **mosaic variegated aneuploidy syndrome type 1 (MVA1)** caused by **biallelic loss of function of `BUB1B`** — compound heterozygous `c.2210T>G` p.(Leu737Ter) and `c.3006T>G` p.(Asn1002Lys), identified in Track 1 and since confirmed by five independent lines of evidence. `BUB1B` encodes **BUBR1**, the pseudokinase that keeps the **mitotic spindle-assembly checkpoint (SAC)** closed until every chromosome is correctly attached. A hypomorphic BUBR1 produces **two consequences from the same defect**, and each opens a separate therapeutic angle:

Consequence of low BUBR1	Phenotype in the record	Therapeutic axis
Chromosome missegregation → constitutional mosaic aneuploidy	rhabdomyosarcoma on the phenotype list; childhood cancers reported in BUB1B-related MVA include embryonal rhabdomyosarcoma and Wilms tumour — risk clearly increased but not quantifiable	exploit the **metabolic / proteotoxic vulnerability of aneuploid cells**
BUBR1 insufficiency → accelerated cellular senescence	severe IUGR and prematurity, failure to thrive, skeletal-muscle atrophy, nephrocalcinosis	**clear accumulating senescent cells** (senolysis)

The disease gene is not a drug target. `BUB1B` is pan-essential (DepMap CRISPR gene effect median −1.21), has **zero drugs or clinical candidates** in Open Targets, and the only compounds near it in the interactome are spindle poisons and cytotoxic chemotherapy — which would weaken an already weak checkpoint. The opportunity is **downstream** (§3).

Proposed agents — ordered by the strength of the clinical evidence in children (the mechanistic rationale is strong for all three):

#	Agent	Role	Dose and its source	Evidence in children	Confidence
1	**Metformin**	AMPK activation → metabolic stress that aneuploid cells tolerate poorly	start 500 mg/day, +500 mg/day to 500 mg b.i.d. (<10 y) or 1000 mg b.i.d. (≥10 y) — protocol of NCT03398824	Fanconi anaemia (a paediatric genome-instability syndrome), 6–35 y, median 9.4 y, 14 treated: no metabolic acidosis, no grade ≥3 events, haematological response 4/13	**highest**
2	**Nicotinamide riboside (NR)**	NAD^+ → SIRT2 → BUBR1: upstream adjuvant that could raise residual BUBR1	25 mg/kg/day — ataxia-telangiectasia study	ataxia-telangiectasia, 24 patients (17 children), 4 months: tolerated well; improved ataxia scores and IgG. **None in cancer predisposition**	moderate (safe; the MVA mechanism is pre-clinical)
3	**Fisetin**	senolytic — the progeroid axis	20 mg/kg/day × 2 consecutive days, every 4 weeks — an **adult** regimen, extrapolated by weight	**none.** Adult trials only (SEN-SURVIVORS ≥18 y; older adults ≥60 y)	lowest clinically; strongest mechanistic fit to the progeroid axis

Regulatory status: metformin is an approved medicine. NR and fisetin are marketed as dietary supplements, not approved medicines; any use needs a pharmaceutical-grade source and clinical supervision (§9).

Context, not repurposing: intensive tumour surveillance (renal ultrasound every 3 months from birth to age 7, as recommended for all MVA types) is the highest-impact recommended intervention and is independent of everything above. Low-dose aspirin ± resveratrol are an optional low-evidence adjunct (adult evidence only; Reye-syndrome precautions).

What this is: a mechanistically grounded hypothesis, cross-checked by eleven lines of evidence, for an n-of-1 discussion between clinical genetics and paediatric oncology. There has never been an interventional trial in MVA1 (§7). It is **not** a proven treatment (§9).

What is new here: (i) the two-axes-from-one-defect framing, anchored in the BubR1-hypomorphic mouse in which clearing senescent cells was first shown to delay age-related disease; (ii) a **druggability gate** showing the checkpoint is intractable, which moves the search downstream; (iii) an **evidence ladder** that keeps mechanistic plausibility separate from paediatric clinical evidence; (iv) a proposed **n-of-1 monitoring frame** with pharmacodynamic biomarkers read off primary transcriptome data (§8); (v) everything reproducible from public data on a dual-core laptop (§10–11).

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2. The variant and the broken pathway

2.1 The two alleles

	Allele A	Allele B
GRCh38	chr15:40,209,701 T>G	chr15:40,220,612 T>G
HGVS (NM_001211.6)	c.2210T>G p.(Leu737Ter)	c.3006T>G p.(Asn1002Lys)
Effect	nonsense, exon 17/23 → truncation (null)	missense, exon 23/23, C-terminal pseudokinase domain
Evidence	ClinVar Pathogenic (VCV000533901); gnomAD 7.9×10^{-5}	absent from gnomAD; SIFT 0.01, PolyPhen-2 0.997, AlphaMissense 0.923, CADD 24.5

Both calls are supported by five independent lines: the consortium GATK VCF plus four analyses of the raw reads (read counting after k-mer extraction; bwa-mem + bcftools; bwa-mem + Clair3 targeted; bwa-mem + Clair3 whole-genome). Three genome-wide scans (ClinVar on the provided VCF; a novel-variant scan of a 235-gene panel; ClinVar on our independent callset) find **no other gene with two rare deleterious alleles**. The phase of the two alleles (trans) is not proven — there is no parental data (Track 1 caveat).

2.2 A null plus a hypomorph

BUBR1 is a component of the **mitotic checkpoint complex** (BUBR1–BUB3–MAD2–CDC20), which inhibits the anaphase-promoting complex until all kinetochores are attached. The two alleles are different in kind. **A** truncates the protein. **B** replaces Asn1002 — which, in the AlphaFold model of BUBR1 (pLDDT 91 at this position), is **buried in the hydrophobic core of the pseudokinase C-lobe** (161 heavy atoms within 10 Å of its Cα, flanked by Trp973/Trp978 and Phe954/977/996/997) and is ~19 Å from the HRD-like (HGD 880–882) and ~27 Å from the DFG-like (DFS 911–913) motifs, i.e. not in the (pseudo)active site (`investigacion/estructura/n1002_structure.py`). An Asn→Lys change puts a charge in a hydrophobic core: a **folding defect, not a catalytic one** — a **hypomorph**. That is the allele pair expected in MVA1: complete loss of BubR1 is embryonic-lethal in mice (Kim et al. 2005), so a viable patient is expected to carry at least one hypomorphic allele.

2.3 Two consequences, two axes

1. **Weak checkpoint → missegregation → mosaic aneuploidy → tumour predisposition.** The cellular hallmark of BUB1B-related MVA is premature chromatid separation. Cancer risk is clearly increased, but the rarity of the syndrome prevents an accurate estimate (Nakano et al. 2024); the

reported tumours are childhood cancers, including embryonal rhabdomyosarcoma (Hanks et al. 2004) and bilateral Wilms tumour (Ochiai et al. 2019).

2. **BUBR1 insufficiency → senescence → progeroid features.** Mice with low BubR1 levels develop progressive aneuploidy and progeroid features (Baker et al. 2004); p16Ink4a-positive senescent cells accumulate and drive progeroid pathology — including sarcopenia — and their genetic clearance delays it (Baker et al., Nature 2011). That experiment — the first to show that removing senescent cells delays age-related disease — was done in a BubR1 hypomorph.

3. Why repurposing — and why downstream

No cause-directed therapy exists. Management is symptomatic: tumour surveillance, standard protocol treatment of any tumour (genome-instability disorders as a group can be unusually sensitive to radiation and chemotherapy; Nakano et al. 2024), and developmental support.

3.1 The checkpoint is not a drug target (C10)

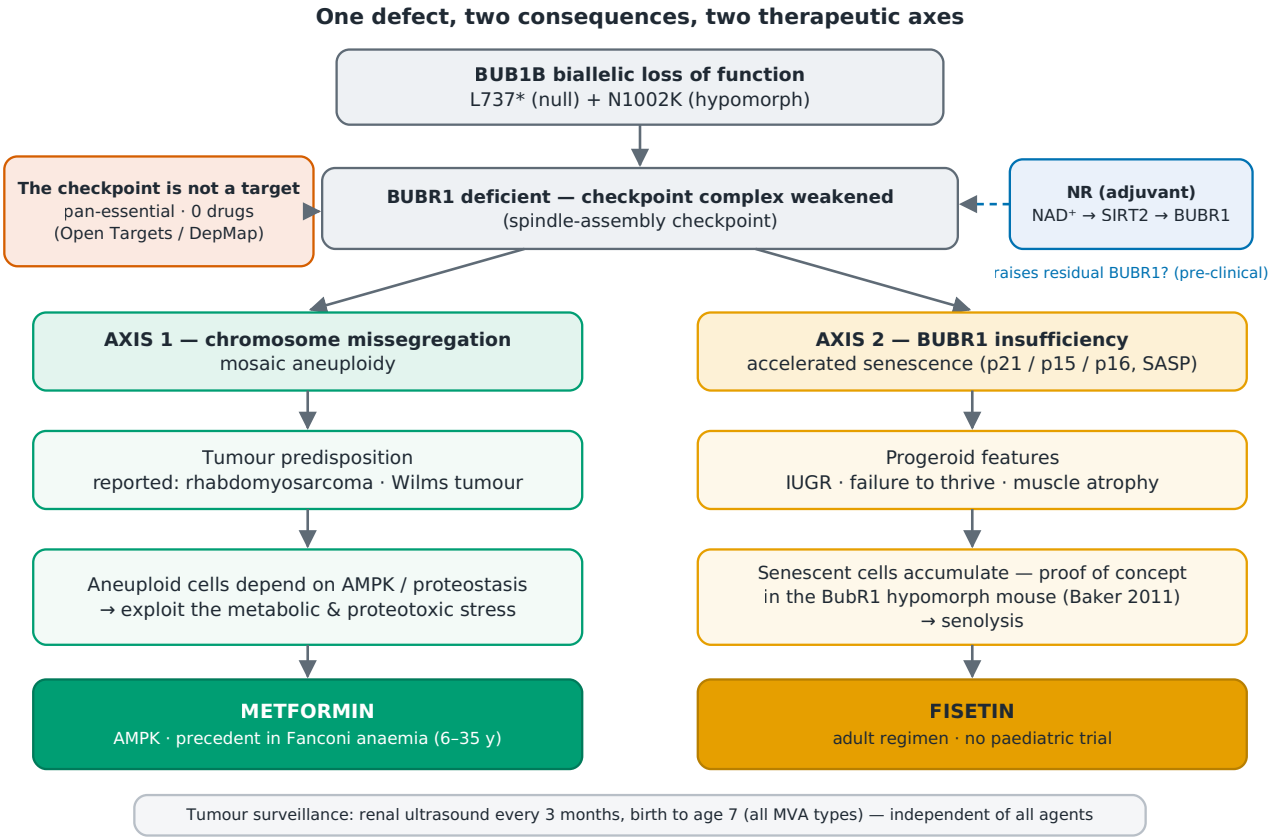


Figure 1 — one defect, two axes, and where the drug space is

Gene	Open Targets "essential" flag (DepMap median gene effect)	Drugs / clinical candidates in Open Targets
BUB1B	**yes (−1.21)**	**0**
BUB1	yes (−0.64)	0
MAD2L1	yes (−1.37)	0
CDC20	yes (−2.09)	0
TTK (MPS1)	yes (−0.84)	2
AURKB	yes (−2.20)	18 — cytotoxic Aurora inhibitors

A median gene effect near or below −1 (the usual common-essential benchmark) means most cell lines need the gene: `BUB1B`, `MAD2L1`, `CDC20` and `AURKB` are past it; `BUB1` and `TTK` are flagged essential but less uniformly. Inhibiting the checkpoint is therefore cytotoxic chemotherapy, not targeted therapy — and, in a child whose checkpoint is already weak, the wrong direction. Two network-proximity analyses agree (C1 shortest path; C5 random walk on STRING ≥700 and ≥900, Spearman ρ −0.58 to −0.67 between them — the sign reflects distance versus proximity): the compounds nearest `BUB1B`/the SAC are Aurora/PLK/MPS1 inhibitors, taxanes, vinca alkaloids and antimetabolites.

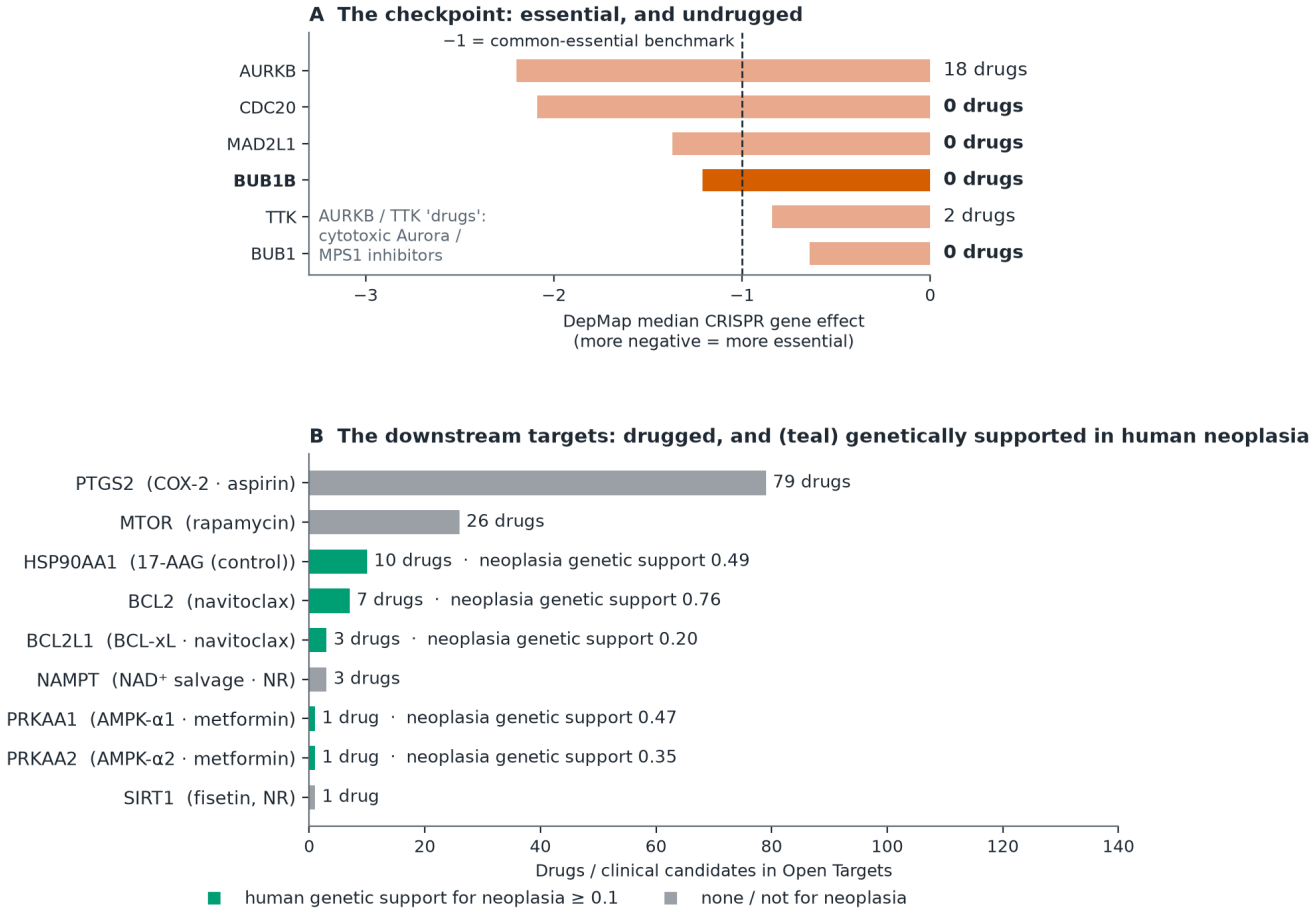


Figure 2 — the checkpoint genes are essential and undrugged; the downstream targets are drugged and genetically supported

3.2 The biology around the gene contains both axes (C6)

The 187-gene STRING neighbourhood of `BUB1B` is, as expected, the checkpoint (Reactome "Cell Cycle" adjusted p = 2.5×10^{-121}; "Mitotic Spindle Checkpoint" 3.4×10^{-119}). The same module is also enriched for **"Cellular Senescence" (p = 3.7×10^{-26}), "SASP" (1.2×10^{-20})** and "Oxidative Stress Induced Senescence" (7.9×10^{-8}), and 37 of its 187 genes belong to **CIN70**, the clinical chromosomal-instability signature (`BUB1B` itself does not). The enrichment was run on the interactome without assuming either axis.

3.3 Primary transcriptome data show both axes (C4)

In GEO GSE277997 (BubR1-insufficient mouse heart; 168 genes up and 160 down at padj below 0.05 and |log2FC| above 0.585), the induced genes include the senescence markers **Cdkn1a (p21), Cdkn2b (p15), Timp1** and **Gdf15**, and the integrated-stress / mitochondrial-stress genes **Fgf21, Trib3** and Gdf15 — all at padj ≤ 1.5×10^{-5} (Gdf15 log2FC 3.4, Fgf21 5.2) — the integrated-stress response characteristic of aneuploid cells (the signature used in C3). **Cdkn2a (p16), Ddit3 (CHOP) and Chac1 pass the threshold only narrowly (padj 0.03–0.05).** A second dataset (GSE134781, Sieben et al. 2020; heterozygous carrier skeletal muscle of knock-ins that mimic patient alleles L1012P — mouse numbering L1002P, about ten residues from the proband's N1002K — and a truncating X753) does **not** reproduce the signature (agreement at chance level), as expected for asymptomatic carriers in a

different tissue; it is used only as context. The deposited table also contains a streak of 165 genes with an almost identical fold-change (log2FC ≈ ±0.81; the vertical columns in Figure 3) — a feature of the table that is not interpreted; the signature-reversal analysis was therefore repeated without them (§4, C4).

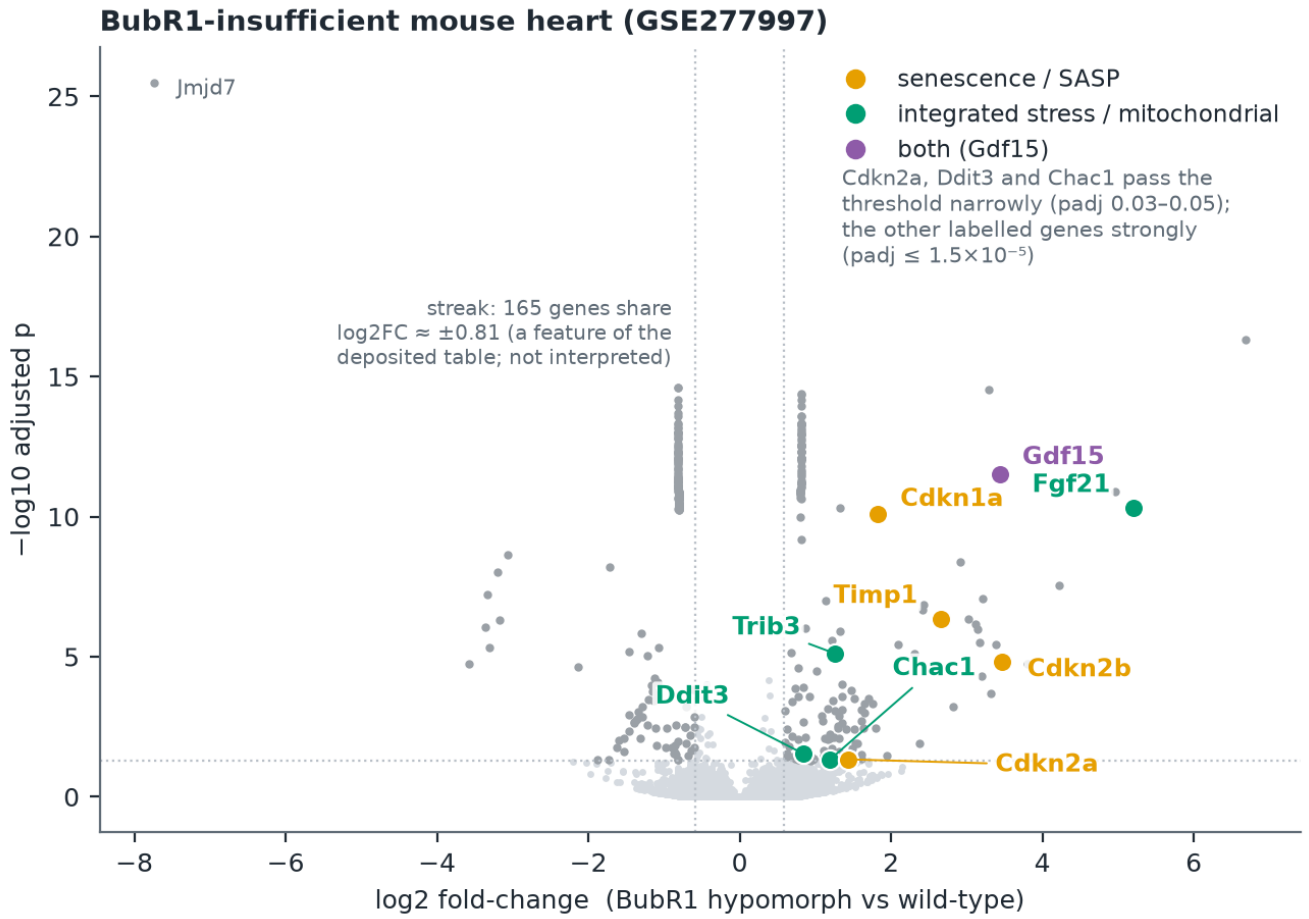


Figure 3 — BubR1-insufficient mouse heart (GSE277997): senescence and integrated-stress genes are induced; the three genes labelled with leader lines pass the threshold narrowly

4. Methods — eleven lines of evidence

Candidate agents were triaged by a curated mechanistic analysis plus ten computational or data-driven analyses (C1–C10). The scripts, outputs and data versions are in the repository (§11).

ID	Question	Data and method	Result
Curated	what do mechanism and literature support?	manual curation over a PubMed / ClinicalTrials.gov evidence pipeline	two axes; senolytics, metformin/AICAR, NAD⁺ precursors; resveratrol + aspirin
C1	which of 8,706 drugs have targets near each disease module?	STRING v12 ≥700 (15,882 genes, 236,712 edges); closest-distance (Guney 2016); z versus 1,000 degree-matched sets; four modules	gene module: spindle/PLK/MPS1 inhibitors and antimetabolites; senescence module: BCL-2/xL (navitoclax z −2.2); aneuploidy-stress module: AICAR −5.7, aspirin −6.0, 17-AAG −4.2, rapamycin −4.0, metformin −3.7
C2	what does full-text literature co-mention with BUB1B/ aneuploidy/ senescence?	Europe PMC, 25 queries, 220 open-access full texts, 6,578-compound lexicon	raw ranking dominated by reagents and chemotherapy; after filtering the curated set re-emerges; two cautions added (TKIs and clonal chromosomal abnormalities; flavonoid combinations)
C3	which perturbagens reverse literature-defined signatures?	L1000CDS2; senescence, aneuploidy-response and CIN70 signatures	HSP90, CDK and mTOR inhibitors reverse the signatures; dasatinib reverses senescence

ID	Question	Data and method	Result
C4	...a signature derived from primary BubR1-insufficiency data?	GEO GSE277997 → L1000CDS2	signature shows both axes (§3.3); reversing mechanism classes (HSP90, mTOR, HDAC, CDK) match C3 and persist when 165 uniform-fold-change genes are excluded (top reverser: the mTORC1/2 inhibitor INK-128); scores low (tissue-level signature)
C5	does C1 survive another metric and network?	personalized PageRank; STRING ≥700 and ≥900	same conclusion; senolytics ~3× closer to the senescence module than to the gene module (mean z 0.60 vs 0.19), mTOR/HSP90/AMPK agents ~2× (1.37 vs 0.64)
C6	what biology surrounds BUB1B?	187-gene interactome module; Enrichr (GO, Reactome, Hallmark)	§3.2
C7	is there clinical precedent?	ClinicalTrials.gov v2: 44 queries, 146 trials seen, 15 read in full with eligibility ages	none in MVA/PCS/BUB1B/CEP57; 5 of 15 include children (§7)
C8	can it be dosed for this child?	trial protocols, labels, PharmCAT v3.4.0 on the proband's variants	§5–6
C9	safety signals?	openFDA FAERS (paediatric strata)	class events as expected; paediatric exposure exists for metformin (523 reports <18 y), sirolimus (672), aspirin (1,344)
C10	is the gene druggable; do the target genes have human genetic support?	Open Targets Platform v4; DepMap	§3.1 and §5

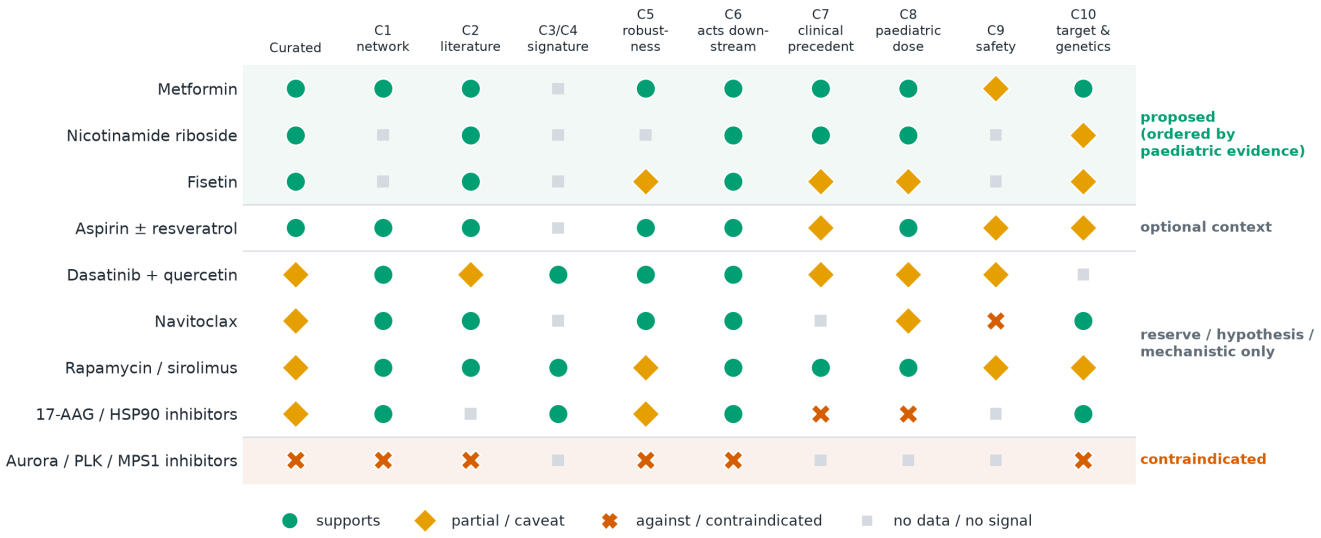


Figure 4 — convergence of the eleven lines of evidence on each candidate. Coding: C5 "supports" = network-proximity z ≥ 2 to a downstream module, "partial" = 1 ≤ z < 2 or not downstream-specific; "no data / no signal" also covers agents that were evaluated but did not emerge (e.g. metformin and fisetin in signature reversal)

What the convergence does and does not show. The methods are independent in data and algorithm, but **not in framing**: C1, C3 and C5 take the two axes as their disease modules. What they test is whether the drugs picked by the curated analysis also emerge from unbiased searches of drug space, and whether anything else does — nothing new, clean and approved did. The framing itself is tested separately by C6 (interactome enrichment that assumes neither axis), C4 (primary transcriptome data), C10 (druggability) and the primary literature.

5. The candidates

5.1 Metformin — best supported

Mechanism. AMPK activation. In the screen of Tang et al. (Cell 2011) the aneuploidy-selective compounds were the energy-stress agent **AICAR** (an AMPK activator), the HSP90 inhibitor 17-AAG and the autophagy inhibitor chloroquine; metformin is an indirect AMPK activator (extrapolation from AICAR). C1: AICAR z −5.7, metformin −3.7. C10: **PRKAA1/PRKAA2 have human genetic support for neoplasia** (Open Targets genetic-association 0.47 gastric carcinoma; 0.35 ovarian neoplasm — direction not indicated).

Evidence. NCT03398824 (Fanconi anaemia, Boston Children's; Pollard et al., Blood Adv 2022): 6–35 years, median 9.4, 14 treated for 6 months on immediate-release metformin dosed as for type 2 diabetes. Starting dose 500 mg/day, increased by 500 mg/day to 500 mg b.i.d. (<10 y) or 1000 mg b.i.d. (≥10 y). Haematological response 4/13 (31 %); **no metabolic acidosis, no hypoglycaemia, no grade ≥3 events attributed to metformin**; diarrhoea in 12/14 (grade 1 in 9, grade 2 in 4); one dose reduction (2000→1500 mg/day). The endpoint was haematological — **not** tumour prevention.

This child. OCT1 (SLC22A1) reduced-function variants R61C, G401S, M420del and G465R are all reference → expected normal hepatic uptake. **Nephrocalcinosis → eGFR at baseline and periodically; hold during dehydration, vomiting or iodinated contrast; B12 yearly.** Lactic acidosis is a rare, labelled class risk tied mainly to renal impairment (it appears among FAERS reports); the Fanconi trial saw none.

5.2 Nicotinamide riboside — safe adjunct

Mechanism. BUBR1 declines with age because NAD^+ falls and SIRT2 can no longer keep BUBR1 lysine-668 deacetylated; overexpressing SIRT2 **or giving the NAD^+ precursor NMN raises BUBR1 levels in mice**, and SIRT2 overexpression extends median lifespan of BubR1-hypomorphic mice (North et al. 2014). NAD^+ repletion is the most upstream intervention — it acts on the cause rather than the consequence — but the evidence is mouse and cell data only, and whether a destabilised BUBR1 (N1002K) responds is unknown.

Evidence. Veenhuis et al. (Mov Disord 2021): open-label proof-of-concept study, 24 patients with ataxia-telangiectasia (17 children), 25 mg/kg/day for 4 months — improved ataxia scores and IgG; "tolerated well"; ataxia worsened again after withdrawal. Further registered trials: NCT04870866 (A-T, ≥3 y), NCT04192136 (Friedreich ataxia, 10–40 y). **No trial in cancer predisposition.** FAERS holds only 36 reports for NR in total; high-dose nicotinamide (not NR) shows hepatic-enzyme signals and inhibits sirtuins — hence NR.

5.3 Fisetin — strongest mechanism, weakest paediatric evidence

Mechanism. Senolysis. The proof of concept is the BubR1 hypomorph itself (§2.3), whose muscle sarcopenia matches the proband's atrophy, and C4 shows p21 and p15 induction (p16 only narrowly) in BubR1-insufficient tissue.

Evidence — adults only. Dose from the Mayo "hit-and-run" trials: 20 mg/kg/day for two consecutive days, repeated monthly. Registered trials: SEN-SURVIVORS (NCT04733534; fisetin or dasatinib + quercetin; **adult survivors ≥18 y** of childhood cancer, n=120) and NCT04313634 (older adults ≥60 y). **No paediatric trial exists; the dose is an adult regimen extrapolated by weight, and paediatric pharmacokinetics are unknown.** Oral bioavailability is low and variable.

This child. Fisetin is reported to inhibit CYP2C9 in vitro (and, like other flavonoids, UGTs and transporters; clinical relevance unknown); the proband is CYP2C9 normal, so the concern is for co-medication (celecoxib, ibuprofen, warfarin, methotrexate, irinotecan), not to the child's own clearance. Why not dasatinib + quercetin: bone growth and development events occurred in 4 % of children in the paediatric dasatinib trial (Gore et al. 2018); clonal chromosomal abnormalities in Philadelphia-negative cells are a recognised phenomenon during TKI therapy (Wang et al. 2013) — a poor fit for a child with constitutional

chromosomal instability; and pleural effusion is the commonest FAERS event for dasatinib. Whether clearing senescent cells helps when senescence is driven by BUBR1 insufficiency — rather than by age or chemotherapy — is untested.

5.4 Reserve, hypothesis-only and excluded

Agent	Status	Reason
Dasatinib + quercetin	reserve	see §5.3; only if fisetin fails and paediatric oncology agrees
Navitoclax	reserve	on-target, dose-limiting thrombocytopenia from BCL-xL inhibition (Mahadevan et al. 2025); platelet-sparing BCL-xL degraders are a future alternative
Rapamycin / sirolimus	hypothesis	strong in C1/C3/C5 and in progeria (NCT02579044, 18 mo–25 y), but immunosuppressive in a child with tumour risk
17-AAG / HSP90 inhibitors	**mechanistic validation only**	strongest computational signal (C1 z −4.2, reverses C3 signatures) but unapproved: 38 oncology trials, abandoned for toxicity
Aurora / PLK / MPS1 inhibitors	**contraindicated**	inhibit an already weak checkpoint
HDAC inhibitors	not repurposing	recurrent in C1/C2/C4; cytotoxic oncology agents
Aspirin ± resveratrol	optional context	chemoprevention evidence only in adults (Lynch syndrome); Reye syndrome — hold in febrile viral illness and peri-operatively

6. Patient-specific considerations

PharmCAT v3.4.0 on the proband's variants (research mode, `--absent-to-ref`; the VCF is variant-only). **Research-grade calls — a clinical decision would need a validated assay** (CYP2D6 is SNV-only; OCT1 was checked by hand).

Gene	Result	Implication
SLC22A1 (OCT1)	no reduced-function variants	metformin uptake expected normal
CYP2C9	*1/*1	fisetin affects co-medication, not this child's clearance
CYP3A4 / CYP3A5	*1/*1 · *3/*3	normal 3A4; non-expressor 3A5
G6PD	normal	aspirin and rasburicase without haemolysis risk
DPYD, TPMT, NUDT15, UGT1A1	normal	standard-dose chemotherapy backbone
CYP2D6	*4/*35 (intermediate)	codeine / tramadol less effective — prefer morphine or hydromorphone
CYP2C19	*2/*17 (intermediate)	voriconazole needs drug monitoring
RYR1	c.6178G>T (VUS)	malignant-hyperthermia susceptibility **uncertain** → prefer TIVA, avoid volatiles and succinylcholine, dantrolene available
F5	rs6025 heterozygous	thromboprophylaxis with central lines / asparaginase

None of these blocks the proposed agents. **The finding that forces action is RYR1, and it concerns anaesthesia, not repurposing** — it belongs here because this child will have many anaesthetics.

Other constraints: nephrocalcinosis (metformin, §5.1); **if a tumour requires chemotherapy, the repurposed agents are paused** (interactions, thrombocytopenia, nephrotoxicity) and resumed in maintenance or remission; fisetin and aspirin are held first.

7. Clinical precedent (C7)

There is **no interventional trial in MVA, premature chromatid separation, BUB1B or CEP57**. The agents have been tested in diseases of the same class, but **only 5 of the 15 trials below enrol children or adolescents, and the senolytic axis is adult-only.** Every row was read in the registry, including the eligibility ages.

Axis	Trial	Population (age)	Phase · N · status
Metformin	**NCT03398824** — Fanconi anaemia	**children/young adults 6–35 y**	2 · 15 · completed, results
Metformin	NCT06519786 — Fanconi cytopenia	**5–18 y**	3 · 30 · status "unknown"
Metformin	NCT01981525 — Li-Fraumeni	adults ≥18 y	1 · 26 · completed
Metformin	NCT02733679 — ataxia-telangiectasia (RAMP)	18–30 y	4 · 27 · completed
NAD^+ (NR)	Veenhuis 2021 (n=24) + NCT04870866 (Phase 2, n=13) + NCT06324877 (ATNAD, n=10) — ataxia-telangiectasia	**children and adults** (Veenhuis: 17 of 24 under 18 y; NCT04870866 ≥3 y; NCT06324877 no minimum age registered)	publication / 2 · active / status "unknown"
NAD^+ (NR)	NCT04192136 — Friedreich ataxia	**10–40 y**	n/a · 75 · completed
NAD^+	NCT05955924 (SPRINTR) — nicotinamide, transplant recipients	adults	3 · 396 · recruiting
mTOR	**NCT02579044** — everolimus + lonafarnib, progeria	**18 months–25 y**	1/2 · 80 · enrolling
mTOR	NCT06950385 (SERENTA) — encapsulated rapamycin, FAP	adults	3 · 168 · recruiting
Senolysis	NCT04733534 (SEN-SURVIVORS) — fisetin or D+Q	**adult** survivors of childhood cancer, ≥18 y	2 · 120 · active
Senolysis	NCT04313634 — senolytics, skeletal health	older adults ≥60 y	2 · 74 · completed
Senolysis	NCT02652052 — D+Q, stem-cell-transplant survivors	adults ≥18 y	1 · 9 · completed
Aspirin	NCT02813824 (AAS-Lynch); NCT02497820 (CaPP3-Israel)	adults	3 · 852 / 1,800
Control	tanespimycin / alvespimycin (HSP90)	oncology	38 trials, 1–2, abandoned for toxicity

Reading. Metformin and NR have trials in genome-instability or neuro-metabolic syndromes that include children; mTOR inhibition has one in paediatric progeria; senolytics and aspirin have adult evidence only. The closest precedent is metformin in Fanconi anaemia.

8. How one would know whether it works — a proposed n-of-1 frame

A starting point for the clinical team, not a protocol.

- **Sequential, single-agent introduction** (metformin → NR → fisetin, if ever), each with a stable baseline and predefined stop rules, so that benefit and toxicity can be attributed.
- **Safety.** eGFR, blood count, liver enzymes and B12 at baseline and at the existing 3-monthly surveillance visits; lactate during intercurrent illness. **Stop or hold** for dehydration, contrast, falling eGFR, chemotherapy cycles, thrombocytopenia (aspirin) or fever (aspirin).
- **Exploratory pharmacodynamic biomarkers** — hypotheses from the primary data, unvalidated in humans: **$CD3^+$ T-cell p16INK4a transcript** (used by SEN-SURVIVORS to select and follow

senescence burden); **serum GDF15 and FGF21**, secreted proteins that are strongly induced (log2FC 3.4 and 5.2) in BubR1-insufficient tissue (C4). A fall under treatment would be consistent with an effect on the senescence / integrated-stress axis; no change would not prove its absence.

- **Clinical endpoints** — growth velocity and weight, muscle function, blood counts (the Fanconi trial's endpoint was haematological response), and the tumour surveillance itself.
- Surveillance (renal ultrasound every 3 months to age 7) continues regardless.

9. Limitations and honest framing

1. **No direct evidence in MVA1.** No interventional trial exists and the syndrome is ultra-rare. This is a hypothesis, not a treatment.
2. **A shared frame.** The convergence is within the two-axis frame (§4). C6 and C4 test the frame, but C4 rests on one informative mouse dataset (heart); the muscle carrier dataset did not replicate it.
3. **Uneven evidence between agents.** Metformin's "aneuploidy-selective" effect comes from cell screens of AICAR, extrapolated; the Fanconi endpoint was haematological, not tumour prevention. NR's link to BUBR1 is pre-clinical. **Fisetin has no paediatric data.**
4. **Method caveats.** Network methods ignore dose, pharmacokinetics and tissue; text mining is noisy; L1000CDS2 scores tissue-level signatures poorly; FAERS is numerator-only and confounded by indication; Open Targets genetic support gives no direction of effect.
5. **Pharmacogenomic calls are research-grade** on a variant-only VCF.
6. **The phase of the two alleles is unproven** (no parental samples) — the Track 1 caveat stands.
7. **Supplements.** Fisetin and NR are supplements: product purity and dose vary, and any use would need a pharmaceutical-grade source and clinical supervision.
8. **Off-label, n-of-1.** Any use requires clinical genetics, paediatric oncology, the family and appropriate oversight.

10. How this generalises (scalability)

The approach does not depend on MVA. For any recessive loss-of-function disorder:

1. **Druggability gate (C10).** If the disease gene is essential and undrugged, search downstream; if it is a normal target, search the gene.
2. **Define disease modules** from the gene's high-confidence interactome and test the biology without assuming the answer (C6).
3. **Triage drug space with orthogonal methods** — network proximity under two metrics (C1, C5), signature reversal from literature-defined and primary signatures (C3, C4), full-text mining (C2).
4. **Filter by evidence in the right population** (C7, with eligibility ages), **dose and pharmacogenomics** (C8), **safety** (C9) and **human genetic support** (C10).
5. **Report mechanism and paediatric evidence as separate axes**, and give the clinical team a monitoring frame (§8).

Every step uses public data and APIs; each script keeps its seed genes and module definitions in a short configuration block at the top, so adapting it to another disease means editing a gene list, not the code. The whole C1–C10 pipeline ran on a dual-core i7-6500U laptop (11.8 GB RAM, spinning disk), at no compute cost; most of the wall-clock time is API latency.

11. Reproducibility

Repository layout: `track2/` (curated pipeline, report, figures, audit), `investigacion/repurposing/` (C1–C10 scripts and outputs; `get_data.sh` downloads the public datasets), `investigacion/estructura/` (structural analysis of N1002K) and `video/` (pitch-video generator). Large public datasets are not committed; `DATA.md` lists URLs, versions and md5 checksums.

- **Data and versions:** STRING v12; DGIdb (retrieved 2026-09-01); Broad Drug Repurposing Hub (2020-03-24); ClinicalTrials.gov API v2 (queried 2026-09-09/10 and 2026-10-07); openFDA FAERS (last updated 2026-07-30); Open Targets Platform v4 (queried 2026-09-10); GEO GSE277997 and GSE134781; Enrichr; L1000CDS2; UniProt O60566/Q9Z1S0; PharmCAT 3.4.0; ClinVar (2026-08-29).
- **Software:** Python 3.12.3, numpy 2.5.2, networkx 3.6.1 (no scipy — PageRank by power iteration); bcftools 1.24.
- **Run:** `python3 investigacion/repurposing/cN_*.py` for each analysis; each writes `cN_RESULTADO.md`.
- **Verification:** `track2/audit_report.py` re-checks every number in this report against its source file and recomputes the structural figures (138 checks, all passing; the 9 pharmacogenomic checks need the private genotype summary and run only on the author's machine); `track2/literature_check.md` records, for every literature claim, the abstract or full-text passage that supports it.

12. AI-assistance disclosure

Analysis, code and drafting were done with **Claude (Anthropic) — Sonnet 5, Sonnet 5.5 and, for the final audit, Opus 5.5 — via Claude Code** (agentic CLI), on a **Claude Pro plan (Anthropic consumer terms)**. Claude wrote and ran the scripts, interpreted outputs and drafted this report under human direction; all account, browser and submission actions were taken by the human author. The plan's model-training setting is declared in the methods description (generative-AI question).

Data handling. The controlled-access files (FASTQ, VCF) were processed only by scripts running on the local machine and were never uploaded to Anthropic. What passed through the model's context was what those scripts printed: aggregate results and about 150 individual genotype calls at targeted loci (the MVA genes, a candidate-gene panel and a few pharmacogenes); no sequencing reads, no complete files and no genome-wide genotype tables. In the Track 2 analyses the only data sent to public services were gene symbols, drug names and public expression signatures (STRING-derived gene lists to Enrichr, mouse GEO signatures to L1000CDS2, identifiers to ClinicalTrials.gov, openFDA and Open Targets); the proband's genotypes were never sent to these services, and pharmacogenomic calling (PharmCAT) ran locally. Variant coordinates, without identifiers, were submitted to the Ensembl VEP REST service during the Track 1 work-up.

Verification. Every quantitative claim is traceable to a script output in the repository. Two verification passes — re-reading the registry entry behind each cited trial, and checking every literature claim against its abstract or full text — corrected about a dozen statements in our working notes before they reached this report (for example: trials described as paediatric that enrol adults, a metformin dose that did not match the protocol, a mouse allele ten residues from the proband's variant, and a cancer-risk figure that the 2024 guideline does not support). The full list is in `track2/literature_check.md`.

13. Author's note

I am not a specialist, and the first report said so. This one asks a harder question than "what is the variant?" — it asks what, if anything, could be done — and the honest answer is that nobody knows, because no one has ever tested anything in this disease. So I tried to keep the same discipline: each claim traceable to a script, each uncertainty stated, and the weakest links named rather than hidden — the fisetin dose is an adult dose, the aneuploidy-selective effect of metformin is borrowed from a cell screen, the phase of the two alleles is unproven. I would rather this report be useful and modest than impressive and wrong.

The child behind `PROBAND01` and their family are the reason it exists. Nothing here identifies them beyond what the challenge provided, and nothing here is advice to them; it is an attempt to give their clinicians a well-organised starting point.

14. Key references

- Hanks S, et al. Constitutional aneuploidy and cancer predisposition caused by biallelic mutations in BUB1B. Nat Genet 2004;36:1159 (PMID 15475955).
- Baker DJ, et al. Clearance of p16Ink4a-positive senescent cells delays ageing-associated disorders. Nature 2011;479:232 (PMID 22048312).
- Tang YC, Williams BR, Siegel JJ, Amon A. Identification of aneuploidy-selective antiproliferation compounds. Cell 2011;144:499 (PMID 21315436).
- North BJ, et al. SIRT2 induces the checkpoint kinase BubR1 to increase lifespan. EMBO J 2014;33:1438 (PMID 24825348).
- Yousefzadeh MJ, et al. Fisetin is a senotherapeutic that extends health and lifespan. EBioMedicine 2018;36:18 (PMID 30279143).
- Pollard JA, et al. Metformin for treatment of cytopenias in children and young adults with Fanconi anemia. Blood Adv 2022;6:3803 (PMID 35500223; NCT03398824).
- Veenhuis SJG, et al. Nicotinamide riboside improves ataxia scores and immunoglobulin levels in ataxia telangiectasia. Mov Disord 2021;36:2951 (PMID 34515380).
- Nelson MR, et al. The support of human genetic evidence for approved drug indications. Nat Genet 2015;47:856 (PMID 26121088).
- Guney E, et al. Network-based in silico drug efficacy screening. Nat Commun 2016;7:10331 (PMID 26831545).
- Sieben CJ, et al. BubR1 allelic effects drive phenotypic heterogeneity in mosaic-variegated aneuploidy progeria syndrome. J Clin Invest 2020;130:171 (PMID 31738183; GEO GSE134781).
- Nakano Y, et al. Update on recommendations for cancer screening and surveillance in children with genomic instability disorders. Clin Cancer Res 2024;30:5009 (PMID 39264246).
- Baker DJ, et al. BubR1 insufficiency causes early onset of aging-associated phenotypes and infertility in mice. Nat Genet 2004;36:744 (PMID 15208629).
- Kim M, et al. Newly identified roles for an old guardian: profound deficiency of the mitotic spindle checkpoint protein BubR1. Cancer Biol Ther 2005;4:164 (PMID 15655361).
- Ochiai K, et al. Long-term remission of bilateral Wilms tumors ... premature separation of chromatids / mosaic variegated aneuploidy syndrome. Pediatr Blood Cancer 2019;66:e27804 (PMID 31081598).
- Gore L, et al. Dasatinib in pediatric patients with chronic myeloid leukemia in chronic phase. J Clin Oncol 2018;36:1330 (PMID 29498925).
- Wang H, et al. Clonal chromosomal abnormalities in Philadelphia-negative cells in CML patients treated with nilotinib. Ann Hematol 2013;92:1625 (PMID 23793947).
- Mahadevan D, et al. First-in-human phase 1 study of DT2216, a selective BCL-xL degrader. J Hematol Oncol 2025;18:98 (PMID 41225624).
- Tools and resources: STRING v12; DepMap; Open Targets Platform; Enrichr; L1000CDS2; PharmCAT; ClinicalTrials.gov; openFDA; UniProt; GEO (GSE277997).